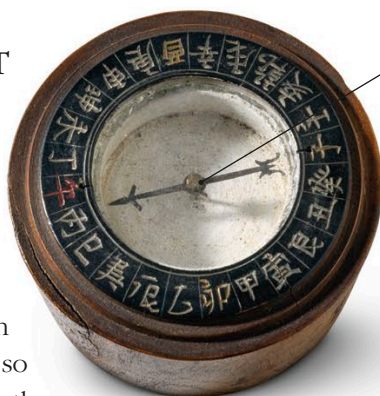


IN THE RIGHT DIRECTION

The compass was invented in China in the 3rd century BCE. Modern instruments have a magnetized needle at the center that works with Earth's magnetic field, so it always points to the north. Early compasses pointed south. From the 11th century CE, sailors began to use the compass at sea.



Chinese compass, mid-19th century

The compass's needle works only if it can spin freely.

The device is held by this ring.

NAVIGATING SEAS

Early mariners calculated their position at sea with a device called an astrolabe that measured the angle between objects in the sky, such as the moon, and the horizon. Ancient Greeks may have designed early astrolabes around 200 BCE and they were definitely in use by the 7th century CE. The astrolabe was then improved by Muslim astronomers who used it to find the direction of Mecca, their holiest city.

Scale to measure the altitude of stars or planets above the horizon

▼ MARINER'S ASTROLABE

Invented in the 16th century CE, this device gave accurate readings on board ships at sea.

Rotating pointer fixed to the center

Viewing hole through which the stars were observed

Heavy bottom to keep the device stable

